

Multiple Choice (10 marks)

1. Which of the following is more appropriate to do feature selection?
 - (a) Ridge
 - (b) Lasso
 - (c) both (a) and (b)
 - (d) neither (a) nor (b)
2. A and B are two events. If $P(A, B)$ decreases while $P(A)$ increases, which of the following is true?
 - (a) $P(A|B)$ decreases
 - (b) $P(B|A)$ decreases
 - (c) $P(B)$ decreases
 - (d) All of above
3. Statement 1| The Stanford Sentiment Treebank contained movie reviews, not book reviews. Statement 2| The Penn Treebank has been used for language modeling.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Let us say that we have computed the gradient of our cost function and stored it in a vector g . What is the cost of one gradient descent update given the gradient?
 - (a) $O(D)$
 - (b) $O(N)$
 - (c) $O(ND)$
 - (d) $O(ND^2)$
5. Which of the following are the spatial clustering algorithms?
 - (a) Partitioning based clustering
 - (b) K-means clustering
 - (c) Grid based clustering
 - (d) All of the above

True / False (10 marks)

Answer True or False

6. True or False: The joint probability distribution $P(X, Y, Z)$ can be expressed as $P(X) * P(Y|X) * P(Z|Y)$.
7. True or False: MLE estimates are always unbiased, which makes them reliable for parameter estimation.
8. True or False: Decision trees are considered unsupervised learning methods in machine learning.
9. True or False: Discriminative approaches model the joint probability $p(y, x)$, which describes the distribution of both input and output variables.
10. True or False: The ResNet-50 model contains over 1 billion parameters, making it one of the largest neural networks available.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to check whether all dictionaries in a list are empty or not.
12. Write a function to check whether the given month number contains 28 days or not.

End of Paper